Day 4: GPUs and Cooling Systems Theory (30 min): Basics of GPUs and cooling systems (air vs. liquid cooling).Practical (30 min):Identify a GPU and cooling system in a PC setup.

1.Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.

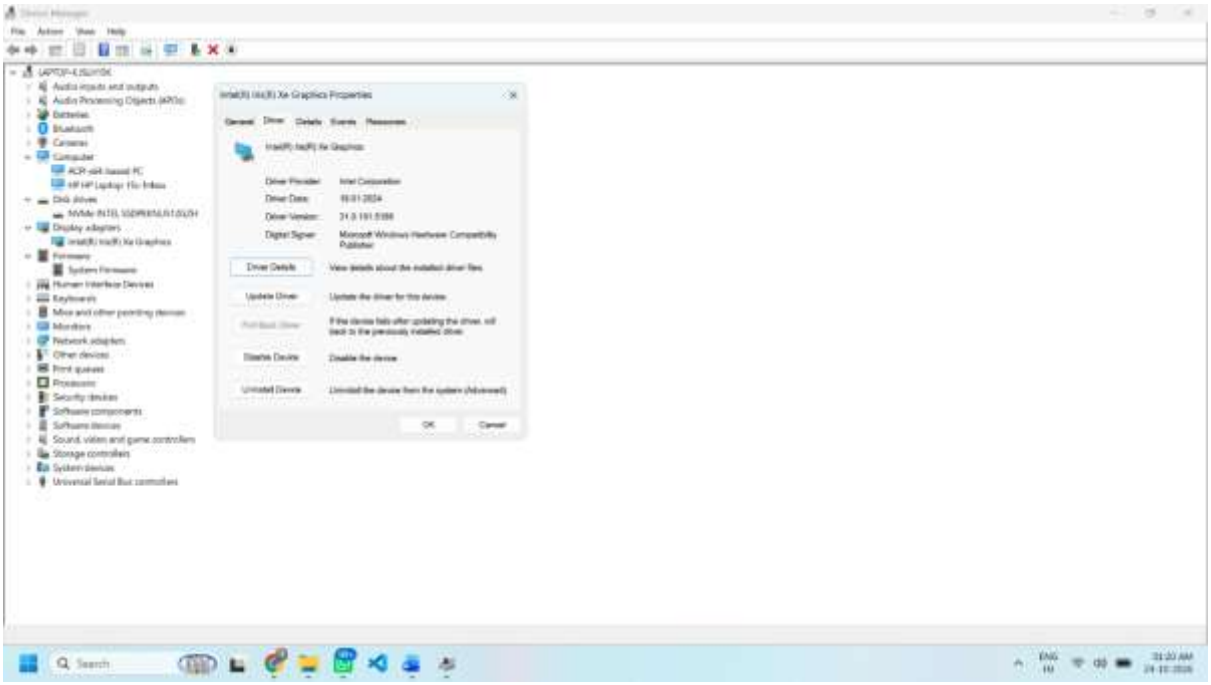
ANS:

STEP 1: Press Windows + X.

STEP 2: Click Device Manager.

STEP 3: Double-click Display adapters.

STEP 4 :



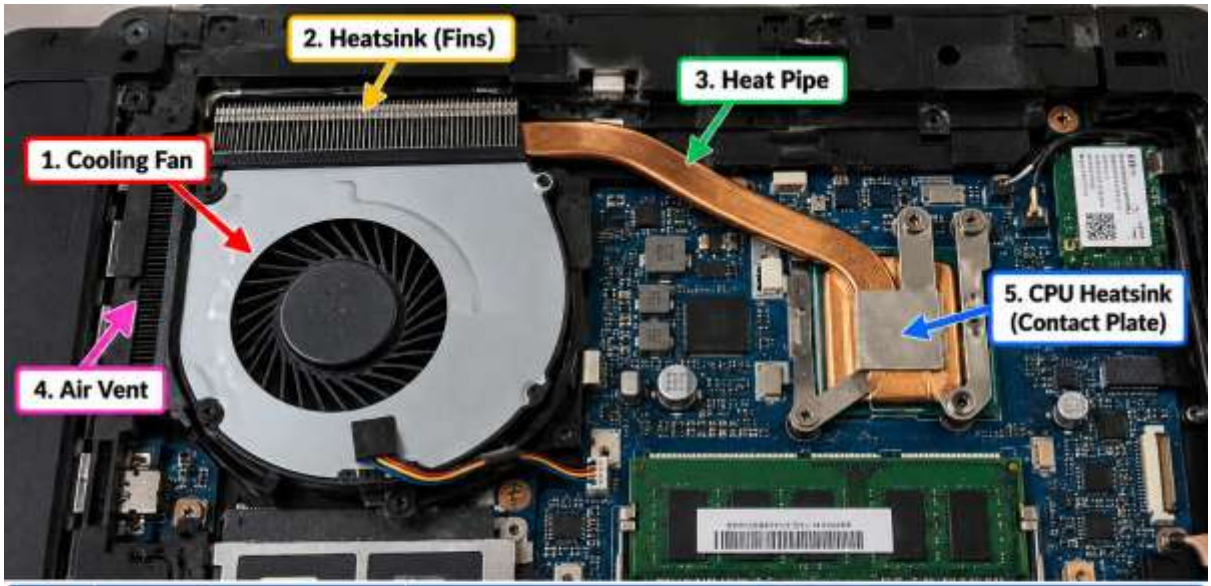
GPU Identification

Detail	Your laptop
GPU Brand	Intel
GPU Model	Intel Iris Xe Graphics
GPU Type	Integrated GPU
Found in	Device Manager → Display adapters

2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.

ANS: Main cooling components and their functions

Component	Function
1. Cooling Fan	Moves air to cool the laptop.
2. Heatsink	Helps remove heat.
3. Heat Pipe	Transfers heat from the CPU to the heatsink.
4. Air Vent	Allows hot air to escape.
5. CPU Contact Plate	Takes heat away from the CPU.



No.	Component	Function
1	Cooling Fan	Pulls in cool air and pushes hot air out through the vent.
2	Heatsink (Fins)	Increases surface area to release heat into the air.
3	Heat Pipe	Transfers heat from the CPU/GPU to the heatsink fins.
4	Air Vent	Lets the hot air out of the laptop.
5	CPU Heatsink (Contact Plate)	Sits on top of the CPU, absorbs heat from the processor and passes it to the heat pipe.

3.Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

ANS:

Feature	Air Cooling	Liquid Cooling
1. Cost	Cheaper	More expensive
2. Cooling Efficiency	Good for normal gaming	Often better for high-performance GPUs
3. Maintenance	Easy to maintain; clean dust regularly	Needs more care; check for leaks in custom loops
4. Noise	Fans can be noisy	Can be quieter, but pumps and fans still make noise
5. Installation	Easy to install	More difficult to install
6. Best Use	Normal computers and gaming	High-end gaming and heavy workloads

4. Imagine you are building a gaming PC for streaming IPL. matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

ANS: My Choice: Liquid Cooling

1. I choose liquid cooling for my gaming PC.
2. It helps keep the GPU cool.
3. It is useful for playing heavy games for a long time.
4. It can make less noise during streaming.
5. It is more expensive than air cooling and needs careful installation.

Why I Choose Liquid Cooling Instead of Air Cooling

1. Liquid cooling can keep the GPU cool during heavy gaming.
2. It is useful for long gaming sessions.
3. It can make less noise than some air coolers.

4. Air cooling is cheaper, but liquid cooling may provide better cooling for high-performance systems.
5. Therefore, I choose liquid cooling for my gaming and streaming PC.